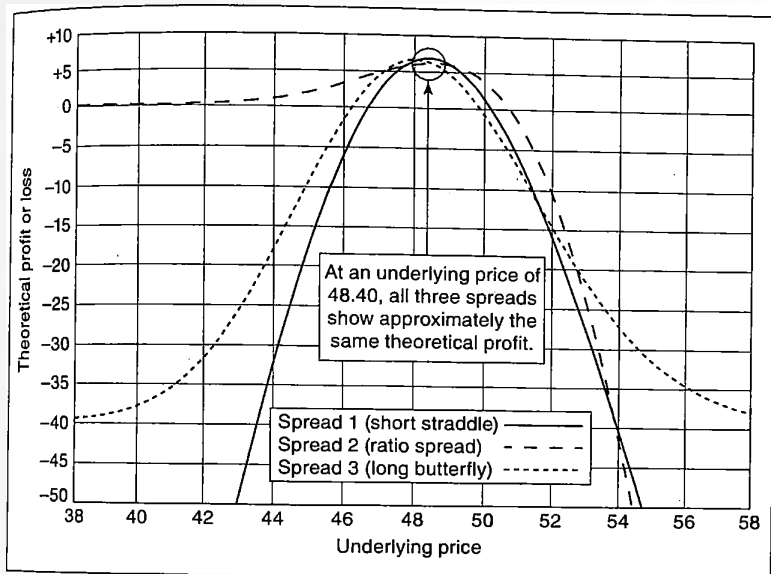


Figure 13-4



Which spread is best? That depends on what the trader is worried about. If the trader is oblivious to the risk, it won't matter which spread he chooses. On average, each position will show a profit of approximately 6.00. If, however, the trader is more worried about a large downward move in the market, then perhaps Spread 2 is best. And if the trader is unwilling to accept the risk of unlimited loss in either direction, then perhaps Spread 3 is best.

In addition to the possibility of a large move, all three positions are exposed to the risk of an incorrect volatility estimate. Because each spread has a negative vega, there will be no problem if, over the life of the option, volatility turns out to be lower than 18 percent. In such a case, the spreads will show a profit greater than originally expected. On the other hand, if volatility turns out to be greater than 18 percent, this could present a problem. What will happen if volatility turns out to be 20 or 25 percent or some higher number? Each spread will be hurt because of the negative vega, but will they be hurt to the same degree?

Because Spread 2 has the smallest vega (-0.875), we might initially conclude that it has the smallest volatility risk. But the vega, like the gamma, changes as market conditions change. If we raise volatility, the vega of Spread 1, the short straddle, will remain essentially unchanged because the vega of an at-the-money option is constant with respect to changes in volatility. But the vega of Spread 3, the long butterfly, will decline because the vega of in-the-money and out-of-the-money options (the May 46 and May 50 puts) will tend to increase as volatility rises. With Spread 2, the vega of both options, the May 50 call and the May 52 call, will begin to increase, so it's not immediately clear what will happen if we increase volatility.